

Xuanzhi Chen

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[Website](#) | [Google Scholar](#) | [GitHub](#) | [LinkedIn](#)

RESEARCH INTERESTS

How gene regulation is organized in tissue, approached from both ends: in situ spatial multi-omics measurement, and the agent-based software infrastructure needed to analyze it reproducibly.

EDUCATION

Peking University

Beijing, China

B.Eng. Candidate in Bioinformatics, School of Life Sciences

Sep. 2023 – Jun. 2027 (Expected)

- **GPA: 3.77/4.0.** Pursuing a double degree in Economics at the National School of Development.
- **Selected coursework:** Computational Immuno-oncology (99/100); Data Structures and Algorithms (98/100); Statistical Analysis of Genomics Data (A); AI Methods and Applications in Life Sciences (A); Fundamentals of Network Science and Systems Biology (A).

PREPRINT

Nianping Liu, **Xuanzhi Chen**, Miao Cui, Xiaoying Liao, Xiaoke Song, Sairam Pantham, Weize Xu, and Xiaojie Qiu.

Bio-Babel: autonomous cross-language reconstruction of agent-ready computational biology ecosystems.

bioRxiv, 2026. [doi:10.64898/2026.08.30.743167](https://doi.org/10.64898/2026.08.30.743167).

RESEARCH EXPERIENCE

College of Future Technology, Peking University

Beijing, China

Undergraduate Researcher; Advisor: Prof. Hu Zeng

Jul. 2024 – Present

- **STATES-GBM:** Built and optimized the upstream spatial omics pipeline for glioblastoma, focusing on cell segmentation and image stitching; performed malignant-cell identification and subtyping and spatial analysis of translation efficiency.
- **iST Agent:** Developed automated contract generation for heterogeneous imaging-based spatial transcriptomics workflows, translating a ~200-file production pipeline into a unified, agent-readable specification covering 47 analysis steps.

Department of Genetics, Stanford University

Stanford, CA

Summer Researcher; Advisor: Prof. Xiaojie Qiu

Jun. 2026 – Present

- **Bio-Babel:** Built an annotation agent that equips reconstructed packages with machine-readable usage contracts; extended coding-agent compatibility and Python-to-C++ reconstruction. Conducted the pancreatic differentiation case study using Bio-Babel's contract-guided analysis agents. Reviewed agent outputs and completed genotype-aware screening and biological interpretation, characterizing reduced maturation signatures despite retained lineage-defining transcription factors under SWI/SNF loss.
Code: [Pancreas case study](#) | [UMI-tools-cpp](#).
- **CycLENS:** Designing a unified model that jointly learns image registration, transcript localization, and barcode decoding from multiplexed fluorescence images, replacing a cascade of independent upstream stages to limit error propagation; architecture implemented and under training.

HONORS AND AWARDS

Peking University Scholarship

2025

Peking University Freshmen Scholarship

2023

Gold Medal, China National Biology Olympiad Final

2022

TECHNICAL SKILLS

Experimental: In situ spatial omics (STATES), molecular cloning, CRISPR-Cas9, PCR/qPCR, Western blotting.

Computational: Python, R; single-cell and spatial omics analysis, biological image analysis, scientific agent development, workflow automation.